

NOTES ON NUMERICAL ANALYSIS

Convergence

Definition: Convergence means that a sequence or series gets closer to a specific value as you add more terms.

- **Example:** Consider the sequence $1, 1/2, 1/3, 1/4, \dots$. As we keep adding terms, the numbers get smaller and closer to 0. So, we say this sequence converges to 0.
- **Why It Matters:** In math, understanding convergence helps us know that we can predict the behavior of sequences and series. If something converges, we can assign it a specific value, which is useful in calculations.

Divergence

Definition: Divergence means that a sequence or series does not settle on any specific value as more terms are added.

- **Example:** Look at the sequence $1, 2, 3, 4, \dots$. This sequence keeps increasing without bound—it goes to infinity. So, we say it diverges.
- **Why It Matters:** Recognizing divergence is important because it indicates that we can't assign a finite value to the series. In calculations, this means the results can be unpredictable.

Accuracy

Definition: Accuracy refers to how close a measured or calculated value is to the true or actual value.

- **Example:** If the actual weight of an object is 5 kg and your measurement is 5.1kg, your measurement is not very accurate because it is far from the true weight.

Precision

Definition: Precision describes how consistent or repeatable measurements are, regardless of whether they are accurate.

- **Example:** If you weigh an object three times and get 5.1 kg each time, your measurements are precise because they are consistent, even if they are not accurate.

Round-off Errors

Definition: Round-off errors happen when numbers are shortened or rounded to fit a certain number of digits, which can cause small mistakes in calculations.

- **Example:** When calculating the value of π , if you use 3.143 instead of 3.141593, the difference introduces a round-off error.

Floating-point Representation

Definition: Floating-point representation is a way computers store real numbers to handle a wide range of values, using a fixed number of digits for the number's significant figures and an exponent.

- **Example:** The number 123.45 might be stored as 1.2345×10^2 in floating-point format. This allows for efficient storage, but it can lead to round-off errors because not all decimal numbers can be represented exactly.

Conditions for Convergence

1. **Continuity:** The function $f(x)$ must be continuous on the closed interval $[a, b]$.
2. **Sign Change:** The function values at the endpoints must have opposite signs:
 - $f(a) \cdot f(b) < 0$.
This indicates that there is at least one root in the interval $[a, b]$ due to the Intermediate Value Theorem.
3. **Interval Halving:** The method effectively reduces the size of the interval containing the root with each iteration, thus converging to the root.

Conditions for Failure (Informal "Divergence")

1. **Lack of Continuity:** If the function is not continuous on $[a, b]$ the bisection method may not find a root.
2. **No Sign Change:** If $f(a)$ and $f(b)$ do not have opposite signs (i.e., both are positive or both are negative), the method will not converge to a root.

3. **Flat Regions or Oscillations:** If the function has flat regions where it does not cross the x-axis, the method may converge very slowly or appear to stagnate.

Qn) Determine the range of integers in base 10 that can be represented on a 32-bit computer.

In a 32-bit computer, integers are typically represented using 32 bits. The range of integers depends on whether the numbers are **signed** (i.e., can be positive or negative) or **unsigned** (i.e., only non-negative).

1. Unsigned 32-bit Integer:

- An unsigned integer uses all 32 bits to represent non-negative values.
- The smallest value is 0.
- The largest value is when all 32 bits are set to 1, which is $2^{32} - 1$.

Range (Unsigned) : 0 to $2^{32} - 1 = 0$ to 4,294,967,295

2. Signed 32-bit Integer:

- A signed integer uses the first bit as a **sign bit** (0 for positive, 1 for negative), leaving 31 bits for the magnitude.
- The smallest value is when the sign bit is 1 and the remaining bits are 0, representing -2^{31} .
- The largest value is when the sign bit is 0 and the remaining 31 bits are all 1, representing $2^{31} - 1$.

Range (Signed) : -2^{31} to $2^{31} - 1 = -2,147,483,648$ to 2,147,483,647

Bisection Method

The **Bisection Method** is a simple root-finding technique that works by repeatedly halving an interval that contains the root. It is based on the idea that if $f(x)$ changes sign between two points, there is at least one root between those points.

Bisection Method Steps

1. Choose Interval:

- Select lower guess x_l and upper guess x_u such that $f(x_l) \cdot f(x_u) < 0$ (indicating a sign change).

2. Estimate Root:

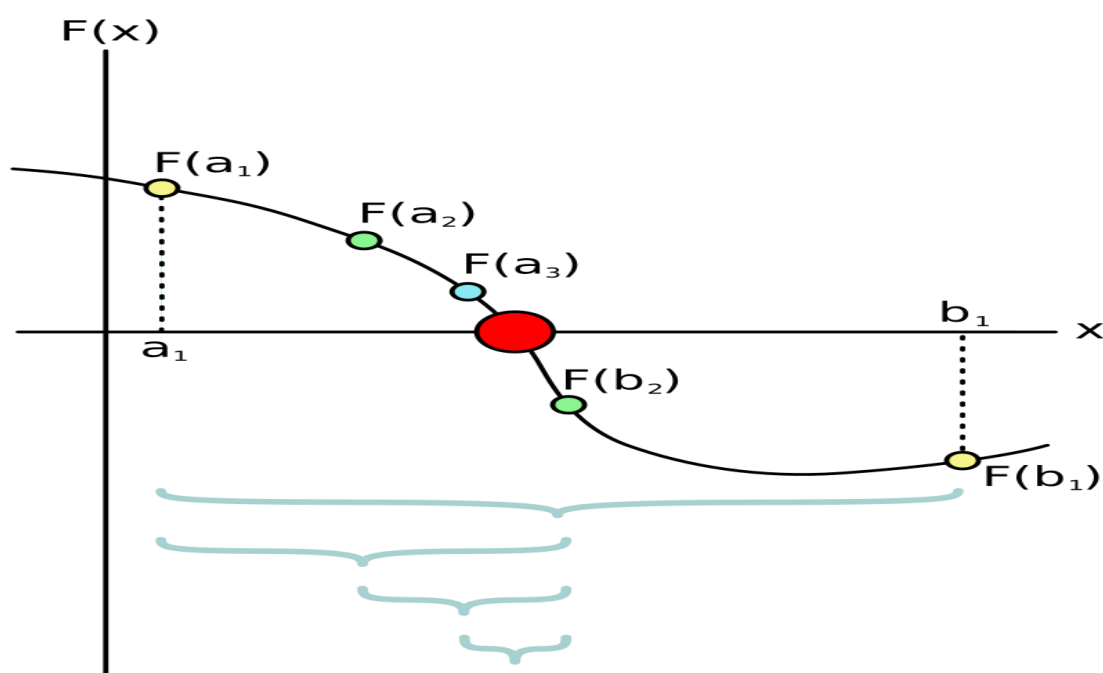
- Calculate the midpoint:

$$x_r = \frac{x_l + x_u}{2}$$

3. Evaluate Subintervals:

- **(a)** If $f(x_l) \cdot f(x_r) < 0$: The root lies in the lower subinterval. Set $x_u = x_r$ and return to Step 2.
- **(b)** If $f(x_l) \cdot f(x_r) > 0$: The root lies in the upper subinterval. Set $x_l = x_r$ and return to Step 2.
- **(c)** If $f(x_l) \cdot f(x_r) = 0$: The root is x_r ; terminate the computation.

Graph-



False Position Method

The False Position Method (or Regula Falsi) is an iterative root-finding technique that combines features of the Bisection Method and linear interpolation. It uses two points to approximate the root and improves the estimate based on the function values.

Steps of the False Position Method

1. Initial Interval Selection:

- Choose two initial guesses, a and b , such that $f(a)$ and $f(b)$ have opposite signs (i.e., $f(a) \cdot f(b) < 0$).

2. Calculate the False Position:

- Compute the point c using the formula:

$$c = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

- This point c is where the line connecting $(a, f(a))$ and $(b, f(b))$ intersects the x -axis.

3. Evaluate the Function:

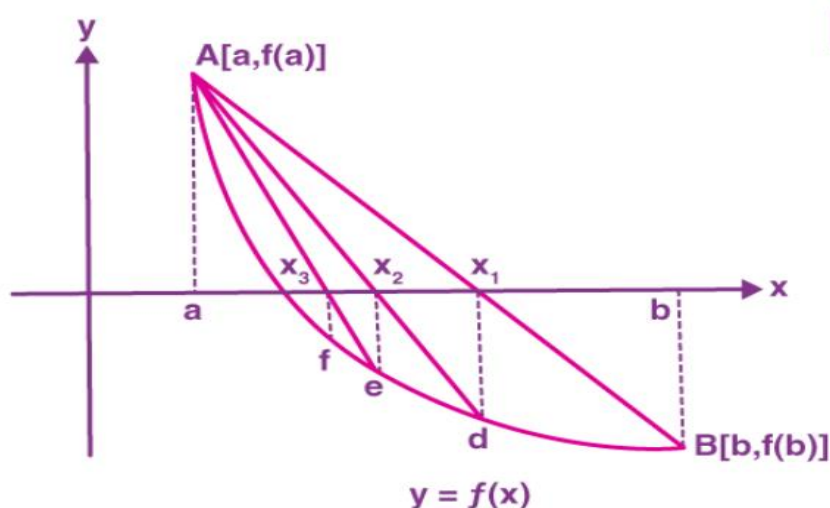
- Calculate $f(c)$.

4. Update the Interval:

- If $f(c) = 0$: c is the root.
- If $f(a) \cdot f(c) < 0$: Set $b = c$ (the root lies in the interval $[a, c]$).
- If $f(b) \cdot f(c) < 0$: Set $a = c$ (the root lies in the interval $[c, b]$).

5. Repeat:

- Repeat steps 2 to 4 until the desired accuracy is achieved, typically when $|f(c)|$ is sufficiently small or the interval $[a, b]$ is sufficiently narrow.



Newton-Raphson Method

The **Newton-Raphson method** is an iterative technique used to find the root of a function $f(x) = 0$. It uses the function's derivative to approximate the root and typically converges very quickly when close to the actual root.

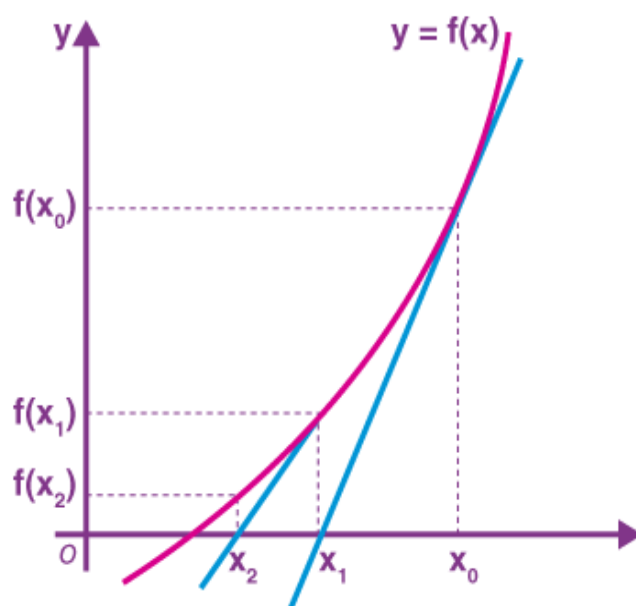
Steps:

1. **Initial Guess:** Start with an initial guess x_0 close to the root.
2. **Iteration Formula:** The next approximation x_{n+1} is calculated using the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Here, $f'(x_n)$ is the derivative of $f(x)$ at x_n .

3. **Repeat:** Continue iterating until the difference between successive guesses is small or the function value is close to zero.
4. **Stopping Condition:** Stop when $|x_{n+1} - x_n|$ is smaller than a tolerance or $|f(x_{n+1})|$ is very close to zero.



Secant Method

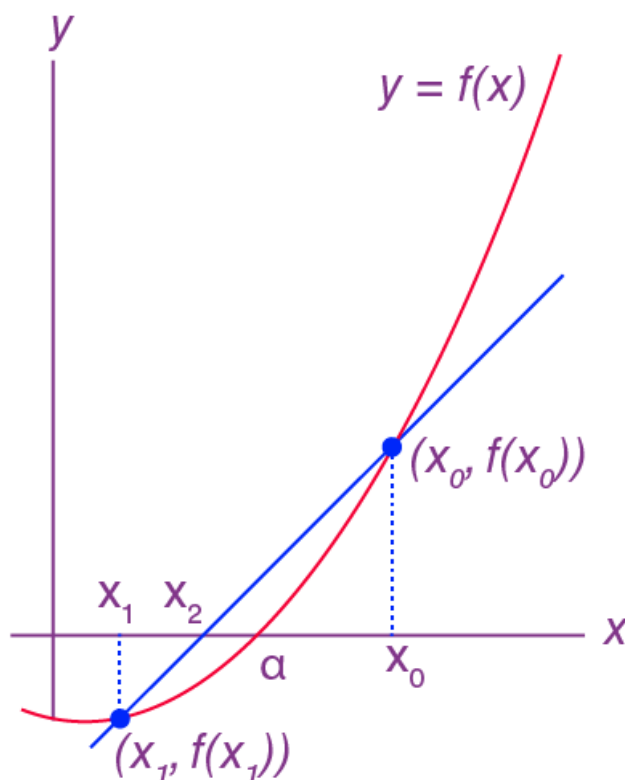
The **secant method** is an iterative technique for finding the root of a function $f(x) = 0$, similar to the Newton-Raphson method but without requiring the derivative.

Steps:

1. **Initial Guesses:** Start with two initial points, x_0 and x_1 , close to the root.
2. **Secant Line:** Use the secant line between these points to approximate the root. The formula is:

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

3. **Iteration:** Repeat using the updated points until the difference between successive guesses is smaller than a tolerance or the function value is close to zero.
4. **Convergence Check:** Stop when $|x_{n+1} - x_n|$ or $|f(x_{n+1})|$ is sufficiently small.



1. Bisection Method Example:

The Bisection Method works by repeatedly halving an interval that contains a root.

Problem: Find the root of $f(x) = x^2 - 4$ between $x = 0$ and $x = 3$.

Steps:

- $f(0) = 0^2 - 4 = -4$ (negative)
- $f(3) = 3^2 - 4 = 5$ (positive)

Start halving the interval:

- Midpoint: $x = \frac{0+3}{2} = 1.5$
- $f(1.5) = 1.5^2 - 4 = -1.75$ (negative)

Next interval is $[1.5, 3]$ because $f(1.5)$ is negative and $f(3)$ is positive. Continue halving the interval until the desired accuracy is reached.

2. False Position Method Example:

The False Position Method improves on the Bisection Method by using a linear approximation between the function values.

Problem: Find the root of $f(x) = x^2 - 4$ between $x = 0$ and $x = 3$.

Steps:

- $f(0) = -4, f(3) = 5$.
- Estimate the root using the formula:

$$x_{\text{new}} = x_1 - \frac{f(x_1)(x_2 - x_1)}{f(x_2) - f(x_1)}$$
$$x_{\text{new}} = 0 - \frac{-4 \times (3 - 0)}{5 - (-4)} = 0 + \frac{12}{9} = 1.33$$

Continue with $[1.33, 3]$ as the next interval.



3. Newton-Raphson Method Example:

This method uses the function's derivative to estimate the root.

Problem: Find the root of $f(x) = x^2 - 2$ starting at $x_0 = 1$.

Steps:

- The derivative of $f(x) = x^2 - 2$ is $f'(x) = 2x$.
- Use the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

For $x_0 = 1$:

$$x_1 = 1 - \frac{1^2 - 2}{2(1)} = 1 - (-1)/2 = 1.5$$

- Continue iterating: $x_2 = 1.5 - \frac{1.5^2 - 2}{2(1.5)} = 1.4167$.
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4. Secant Method Example:

The Secant Method approximates the root using two initial guesses and does not require a derivative.

Problem: Find the root of $f(x) = x^2 - 2$ with $x_0 = 1$ and $x_1 = 2$.

Steps:

- Use the formula:

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

For $x_0 = 1$ and $x_1 = 2$:

$$x_2 = 2 - \frac{(2^2 - 2)(2 - 1)}{(2^2 - 2) - (1^2 - 2)} = 1.333$$

- Continue iterating to get closer to the root.